



Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

Imię i nazwisko / Name, surname	Grzegorz Piotrowski
Dyscyplina kształcenia / Discipline of education	informatyka techniczna i telekomunikacja / information and communication technology
Wydział / Faculty	W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology
Katedra / Department	K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence
Data rozpoczęcia kształcenia / Date of commencement of the doctoral education	01.10.2025
Nr ORCID	0009-0004-6013-0461
Promotor/ promotorzy / Supervisor(s)	dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012
Promotor pomocniczy / Assistant supervisor	—

2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

Improving generalization of deep learning models in real-to-sim-to-real transfer for physical AI.

3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań częściowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:

- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,
- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,



- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)	Zwięzły opis zadania / Brief description of the task
Semestr 1	1. Literature review in the scope of sim-to-real transfer, domain randomization and domain adaptation, neural scene reconstruction (NeRF, 3D Gaussian Splatting) and learning-based robot navigation and manipulation. 2. Identification of the most important research gaps in the generalization of models trained in simulation and in digital twins. 3. Extending theoretical knowledge, including machine learning theory and learning under distribution shift, necessary for further work on the PhD thesis. 4. Participation in the Doctoral School courses.
Semestr 2	1. Continuation of the literature review – participation in the internal reading group, discussing novel methods in sim-to-real transfer and embodied AI. 2. Initial experiments with building digital twins from real captures and training navigation models in them. 3. Specifying the Individual Research Plan. 4. Participation in a machine learning summer school or conference with a poster.



Semestr 3	1. Continuation of the literature review – participation in the internal reading group. 2. Conducting research on which properties of digital twins built from real data determine the generalization of models trained in them. 3. Preparation of the experimental environment (public datasets of real scenes, simulator, reconstruction tools) and applications for computing resources (PLGrid) and a small research grant (SzD Minigrant). 4. Writing a scientific article for a conference from the ministerial list (regarding generalization-aware digital twins).
Semestr 4	1. Continuation of the literature review – participation in the internal reading group. 2. Continuing and finishing the research on generalization-aware digital twins. 3. Research on learning representations invariant to the difference between simulation and reality. 4. Preparation of the Preludium research grant proposal to the National Science Centre. 5. Participation in a scientific conference, preferably with own article.
Semestr 5	1. Continuation of the literature review – participation in the internal reading group. 2. Continuing research on sim-to-real invariant representations, including the use of pretrained foundation models and learned world models. 3. Research on adapting models trained in simulation to reality with a small amount of real data. 4. Validation of the results on a real mobile robot. 5. Writing a scientific article for a conference from the ministerial list (regarding invariant representations and adaptation).
Semestr 6	1. Continuation of the literature review – participation in the internal reading group. 2. Research on the generalization of the developed methods to unseen environments and to another physical task (robotic manipulation). 3. Research internship at a foreign research centre working on robot learning. 4. Writing a scientific article for a conference from the ministerial list (regarding generalization across scenes and tasks).
Semestr 7	1. Continuation of the literature review – participation in the internal reading group. 2. Finishing the studies, final experiments and validation on a real robot, and writing an article for a conference from the ministerial list based on the obtained outcomes. 3. Participation in a scientific conference. 4. Preparing the structure and sketch of the doctoral dissertation.
Semestr 8	Editing the doctoral dissertation, preparation of the final version of the document.



4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission

Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.

September 2029

(rekomendowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /

The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)

5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Deep learning has led to impressive results in computer vision and natural language processing, and it is now increasingly applied to physical AI: robots and other embodied systems that perceive and act in the real world, such as mobile robots navigating buildings or robotic arms manipulating objects. Unlike models trained on images or text, such models need very large amounts of interaction data, which is slow, expensive and sometimes unsafe to collect in the real world. Therefore, they are commonly trained in simulation. Recently, digital twins built from real data became a promising alternative to hand-made simulators: neural scene reconstruction methods, such as Neural Radiance Fields and 3D Gaussian Splatting, turn a short video of a real place into a photorealistic 3D model in which a robot can be trained at scale. This gives the real-to-sim-to-real loop: real data is transferred into a simulation, a model is trained there, and the model is transferred back to the real world.

However, the usefulness of this loop depends entirely on how well the trained models generalize. A model that performs very well in simulation often fails in reality, because a digital twin never reproduces the real world exactly: appearance, lighting, geometry and physical properties differ. Moreover, models often fail again in places or tasks that were not seen during training. This sim-to-real generalization gap remains one of the main obstacles for physical AI. The methods proposed so far, such as domain randomization, rely largely on hand-tuned heuristics, are typically evaluated on a single task and a few scenes, and do not explain which properties of the simulation and of the learned representations actually determine generalization to reality. Therefore, the dissertation aims at developing methods that improve the generalization of deep learning models trained in real-to-sim-to-real loops, with a particular focus on the digital twin itself, the representations learned in it, and the adaptation of models to reality with little real data.

The research is relevant to machine learning in general, as it concerns learning under distribution shift and the generalization of representations, and it could help to understand why models trained on synthetic data succeed or fail on real data. The results have potential applications in service and logistic robotics, where robots could be trained in a digital copy of a new warehouse, hospital or office and work there reliably, as well as in autonomous systems and any other domain where models are trained on simulated or reconstructed data and used on real sensor data.



6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / Outline of the current state of research in the topic of the doctoral dissertation

Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

Deep reinforcement and imitation learning made it possible to train robots for complex tasks such as visual navigation and manipulation. Since such methods require very large amounts of experience, simulation became the standard training environment. Platforms such as Habitat [1] and ManiSkill3 [2] allow training at the scale of billions of steps, and in simulation point-goal navigation has been practically solved [3]. However, the performance of models in simulation does not always predict their performance in the real world [4], and the transfer of models from simulation to reality, known as the sim-to-real problem, remains one of the central challenges of robot learning [5, 6].

The most common approach to the sim-to-real problem is domain randomization, which varies the appearance

[7] or the dynamics [8] of the simulation so that reality appears as one more variation. Other works adapt

the parameters of the simulator to a few real-world rollouts [9]. Interestingly, the fidelity of the simulator alone does not guarantee better transfer, as lower-fidelity simulation can transfer better in navigation [10]. From the machine learning perspective, the sim-to-real problem is an instance of learning under distribution shift. Domain adaptation theory bounds the error on the target domain by the error on the source domain and a discrepancy between the two domains [11], and domain-adversarial training learns representations that are invariant across domains [12]. Nevertheless, such principled approaches are rarely applied to models trained in simulation for physical tasks, where randomization heuristics dominate.

A major recent development is the construction of simulations directly from real data. Neural Radiance Fields [13] and 3D Gaussian Splatting [14] reconstruct photorealistic 3D scenes from ordinary images, and datasets such as ScanNet++ [15] provide real indoor scenes captured both with high-end and consumer devices. Such digital twins have been used to train navigation models from a phone capture of a room [16] or from monocular videos of urban scenes [17], and robotic manipulation policies from a scan of the target scene [18], showing that real-to-sim-to-real learning improves real-world performance in comparison with generic simulators. The physical parameters of a digital twin can be estimated from a small amount of real data [19]. However, these works treat the digital twin mostly as a fixed, as-accurate-as-possible copy of reality. It is not known which properties of a twin (appearance, geometry, lighting or physics) actually determine the generalization of the trained models, and the twins are randomized uniformly rather than according to where they are uncertain or where it matters for the task. Moreover, the evaluation is usually limited to a handful of scenes and a single task.

In parallel, pretrained foundation models changed how robot policies are built. General navigation models trained on data from many robots [20, 21] and vision-language-action models [22] transfer to new robots and environments, and self-supervised visual encoders [23] provide representations that are robust to many appearance changes. Learned world models predict future observations and can be used for planning in navigation [24]. These models are a promising basis for generalization, but it has not been studied how their representations behave under the specific shift between a digital twin and reality, and how they should be trained in a twin so that this shift does not harm their performance. Finally, adapting a model trained in simulation to reality usually relies on real data



collected at random or by hand, and on co-training with simulated data [25], without selecting the real data that is most informative about the remaining gap.

In summary, the literature shows that real-to-sim-to-real learning is effective, but the generalization of the trained models to reality, to unseen scenes and to other tasks is still limited and poorly understood. The proposed dissertation addresses this gap by studying the generalization of deep learning models at each stage of the real-to-sim-to-real loop.

References

- [1] Savva, M., Kadian, A., Maksymets, O., et al. (2019). Habitat: A platform for embodied AI research. ICCV.
- [2] Tao, S., et al. (2025). ManiSkill3: GPU parallelized robotics simulation and rendering for generalizable embodied AI. RSS.
- [3] Wijmans, E., et al. (2020). DD-PPO: Learning near-perfect PointGoal navigators from 2.5 billion frames. ICLR.
- [4] Kadian, A., et al. (2020). Sim2Real predictivity: Does evaluation in simulation predict real-world performance? IEEE Robotics and Automation Letters, 5(4).
- [5] Zhao, W., Queralta, J. P., & Westerlund, T. (2020). Sim-to-real transfer in deep reinforcement learning for robotics: a survey. IEEE SSCI.
- [6] Höfer, S., et al. (2021). Sim2Real in robotics and automation: Applications and challenges. IEEE Transactions on Automation Science and Engineering, 18(2).
- [7] Tobin, J., et al. (2017). Domain randomization for transferring deep neural networks from simulation to the real world. IROS.
- [8] Peng, X. B., Andrychowicz, M., Zaremba, W., & Abbeel, P. (2018). Sim-to-real transfer of robotic control with dynamics randomization. ICRA.
- [9] Chebotar, Y., et al. (2019). Closing the sim-to-real loop: Adapting simulation randomization with real world experience. ICRA.
- [10] Truong, J., et al. (2022). Rethinking sim2real: Lower fidelity simulation leads to higher sim2real transfer in navigation. CoRL.
- [11] Ben-David, S., et al. (2010). A theory of learning from different domains. Machine Learning, 79.
- [12] Ganin, Y., et al. (2016). Domain-adversarial training of neural networks. Journal of Machine Learning Research, 17.
- [13] Mildenhall, B., et al. (2020). NeRF: Representing scenes as neural radiance fields for view synthesis. ECCV.
- [14] Kerbl, B., Kopanas, G., Leimkühler, T., & Drettakis, G. (2023). 3D Gaussian splatting for real-time radiance field rendering. ACM Transactions on Graphics, 42(4).
- [15] Yeshwanth, C., Liu, Y.-C., Nießner, M., & Dai, A. (2023). ScanNet++: A high-fidelity dataset of 3D indoor scenes. ICCV.
- [16] Chhablani, G., et al. (2025). EmbodiedSplat. ICCV. arXiv:2509.17430.
- [17] Xie, Z., et al. (2025). Vid2Sim: Realistic and interactive simulation from video for urban navigation. CVPR.
- [18] Torne, M., et al. (2024). Reconciling reality through simulation: A real-to-sim-to-real approach for robust manipulation. RSS.
- [19] Ramos, F., Possas, R., & Fox, D. (2019). BayesSim: Adaptive domain randomization via probabilistic inference for robotics simulators. RSS.
- [20] Shah, D., et al. (2023). GNM: A general navigation model to drive any robot. ICRA.
- [21] Shah, D., et al. (2023). ViNT: A foundation model for visual navigation. CoRL.
- [22] Kim, M. J., et al. (2024). OpenVLA: An open-source vision-language-action model. CoRL.
- [23] Oquab, M., et al. (2024). DINOv2: Learning robust visual features without supervision. TMLR.
- [24] Bar, A., et al. (2025). Navigation world models. CVPR.
- [25] Maddukuri, A., et al. (2025). Sim-and-real co-training: A simple recipe for vision-based robotic manipulation. RSS.

7. Pytania i hipotezy badawcze / Research questions and hypotheses



Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).

The research carried out in the proposed doctoral dissertation is aimed at studying the following research questions.

1. **Which properties of a digital twin built from real data determine the generalization of models trained in it?** A digital twin reproduces the appearance, geometry, lighting and physical properties of a real scene only approximately, and the errors differ between regions of the scene. It is hypothesized that not all of these errors are equally harmful, and that a model generalizes better to reality when the twin is accurate and varied where it matters for the task and where the reconstruction is uncertain, rather than randomized uniformly. The research should identify these properties and develop a method for building and varying digital twins accordingly.
2. **How to learn representations that are invariant to the difference between simulation and reality?** Models trained in a digital twin tend to exploit features that do not exist in reality, such as reconstruction artifacts. Pretrained foundation models and self-supervised encoders provide robust representations, but it is not known how they behave under the shift between a twin and reality. It is hypothesized that training objectives which enforce invariance between paired observations from the twin and from reality, possibly supported by learned world models, reduce the generalization gap. The research should also examine where in the model this gap arises.
3. **How to adapt a model trained in simulation to reality using a small amount of real data?** Some difference between the twin and reality always remains. It is hypothesized that selecting the real data where the twin and reality disagree the most allows the model, and the twin itself, to be corrected with much less real data than random selection.
4. **Do the improvements generalize to unseen environments and to other physical tasks?** A valuable method should not be tied to a single scene or task. Therefore, the developed methods shall be examined on environments not seen during training and on two different physical tasks, robot navigation and robotic manipulation, to verify that the improvements in generalization are general.

8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).

The proposed Ph.D. dissertation aims to introduce several contributions to the development of the scientific discipline. First and foremost, the research is intended to provide new methods that improve the generalization of deep learning models trained in simulations built from real data, which is one of the main obstacles in applying machine learning to physical AI. The methods are intended to be general, i.e., not tied to a particular robot, scene or task, and are planned to be evaluated on robot navigation and robotic manipulation.

Furthermore, the research seeks to provide a better understanding of the sim-to-real generalization gap: which properties of a digital twin matter for the generalization of the trained model, and where in the model the gap arises. Such knowledge could introduce new perspectives for the design of simulations and of training procedures, and could help to build more principled and interpretable approaches than the currently dominating randomization heuristics. Since the sim-to-real problem is an instance of learning under distribution shift, the findings are also expected to be relevant to



domain adaptation, representation learning and the adaptation of pretrained foundation models in general.

Finally, the developed methods for adapting models to reality with a small amount of real data could reduce the cost of deploying learning-based robots in new environments. The source code and the experimental setups are planned to be released, which could facilitate further research in the field and attract other researchers. Apart from purely scientific results, the methods could be used in real-world applications such as service, logistic and inspection robotics.

9. Planowane metody badawcze / Planned research methods

Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

The research will follow the methods commonly adopted in machine learning and robot learning projects. Since there is no mathematically proven practice for achieving generalization from simulation to reality, the dominant part of the research will be based on empirical methods. The proposed methods will be described theoretically, and their properties or bounds will be analysed whenever possible, in particular with the use of domain adaptation theory.

Digital twins will be built from real captures of indoor scenes using neural scene reconstruction methods (3D Gaussian Splatting and related methods) and structure-from-motion tools, and imported into robot simulators (e.g. Habitat for navigation and ManiSkill3 for manipulation). Experiments will be conducted mainly on public datasets of real scenes, such as ScanNet++ and MuSHRoom, which contain captures of the same scenes made with different devices. This allows building the twin from one capture and using a separate, higher-fidelity capture as a reference for "reality", so that the sim-to-real gap can be studied in a controlled way on many scenes without an own robot. Navigation and manipulation models will be trained with reinforcement and imitation learning, starting from pretrained foundation models whenever possible. The results will be validated on a real mobile robot in rooms at the university, in cooperation with the robotics laboratories of Wrocław University of Science and Technology.

The quantitative evaluation will rely on standard metrics for embodied tasks, such as success rate and success weighted by path length, and on the generalization gap, i.e., the difference between the performance in simulation and in reality. Particular attention will be paid to generalization, so the evaluation will separate the scenes used for training from unseen scenes, and will include a second physical task. Statistical methods will be leveraged to provide reliable outcomes, including repeated experiments with different random seeds and statistical tests for comparisons with baseline methods. Qualitative evaluation will focus on analysing the learned representations, e.g. by probing, visualizing the latent space and analysing failure cases.

The implementation of experiments will be done mainly in the Python programming language with the PyTorch deep learning library. Source code will be managed with the Git version control system, and experiments will be tracked to provide reproducible results. Due to the high computational demand, the experiments will run on GPU clusters provided by the Department and on the Polish computing infrastructure (WCSS, PLGrid). The work will be partly conducted in cooperation with other Ph.D. students and senior researchers. Code and data are supposed to be opened whenever possible, following open science standards and the licences of the datasets used.

The findings will be published at leading machine learning, computer vision and robotics conferences from the ministerial list, such as NeurIPS, ICML, ICLR, CVPR and RSS. In order to further promote the work and establish new collaborations, it is planned to present results at smaller conferences and workshops, like MLinPL, and at summer schools.

10. Streszczenie popularnonaukowe / Abstract for general public



Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).

Streszczenie popularnonaukowe

Sztuczna inteligencja coraz częściej wychodzi poza ekrany komputerów i trafia do robotów, które poruszają się po budynkach albo przenoszą przedmioty. Aby się tego nauczyć, roboty potrzebują ogromnej liczby przykładów, których zebranie w prawdziwym świecie jest powolne i kosztowne. Dlatego coraz częściej uczą się w symulacji, a w szczególności w cyfrowych bliźniakach, czyli wiernych wirtualnych kopiach prawdziwych miejsc, które można odtworzyć z krótkiego nagrania. Taka kopia nigdy nie jest jednak idealna i model, który w symulacji działa bardzo dobrze, w prawdziwym świecie często popełnia błędy, zwłaszcza w miejscach, których wcześniej nie widział. Celem rozprawy jest opracowanie metod, dzięki którym modele uczone w cyfrowych bliźniakach będą lepiej działać w rzeczywistości. Badania obejmą sprawdzenie, które cechy wirtualnej kopii są naprawdę ważne, nauczanie modeli pomijania różnic między kopią a rzeczywistością oraz dostosowanie modeli do rzeczywistości przy użyciu niewielu prawdziwych przykładów. Metody zostaną sprawdzone na nawigacji robotów mobilnych i na robotach przenoszących przedmioty. Oczekuje się, że pozwolą one w przyszłości szybciej i taniej uczyć roboty pracy w nowych miejscach, takich jak magazyny, szpitale czy biura.

Abstract for general public

Artificial intelligence is increasingly moving beyond computer screens into robots that move around buildings or handle objects. To learn this, robots need a huge number of examples, which are slow and costly to collect in the real world. That is why they increasingly learn in simulation, and in particular in digital twins, that is, faithful virtual copies of real places that can be rebuilt from a short recording. Such a copy is never perfect, however, and a model that works very well in simulation often makes mistakes in the real world, especially in places it has not seen before. The goal of the dissertation is to develop methods that make models trained in digital twins work better in reality. The research will cover finding out which features of the virtual copy really matter, teaching models to ignore the differences between the copy and reality, and adapting models to reality using only a few real examples. The methods will be tested on the navigation of mobile robots and on robots that handle objects. They are expected to make it faster and cheaper in the future to teach robots to work in new places, such as warehouses, hospitals or offices.

11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography

Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of



publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.

May 2027

12. Inne / Other comments

Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).

13. Określenie planowanej formy współpracy z promotorem / Planned form of cooperation with the supervisor

Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.

It is planned to have weekly meetings with the supervisor. The meetings are planned to consist of results presentations, discussions, reviews and joint work on papers. The scope of meetings is also supposed to include the evaluation of progress and the planning of further activities. Moreover, it is planned to discuss results, issues and recent advancements in the field with other Ph.D. students weekly, with the supervisor's participation. Day-to-day communication will take place by e-mail and instant messaging.

14. Opinia promotora pomocniczego / Opinion of assistant supervisor

Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.

nie dotyczy / not applicable

.....
Podpis promotora pomocniczego /
Signature of assistant supervisor

15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....
Data / Date

.....
Podpis doktoranta / Signature of doctoral student

.....
Podpis promotora / Signature of supervisor



.....
Podpis drugiego promotora / [Signature of second supervisor](#)